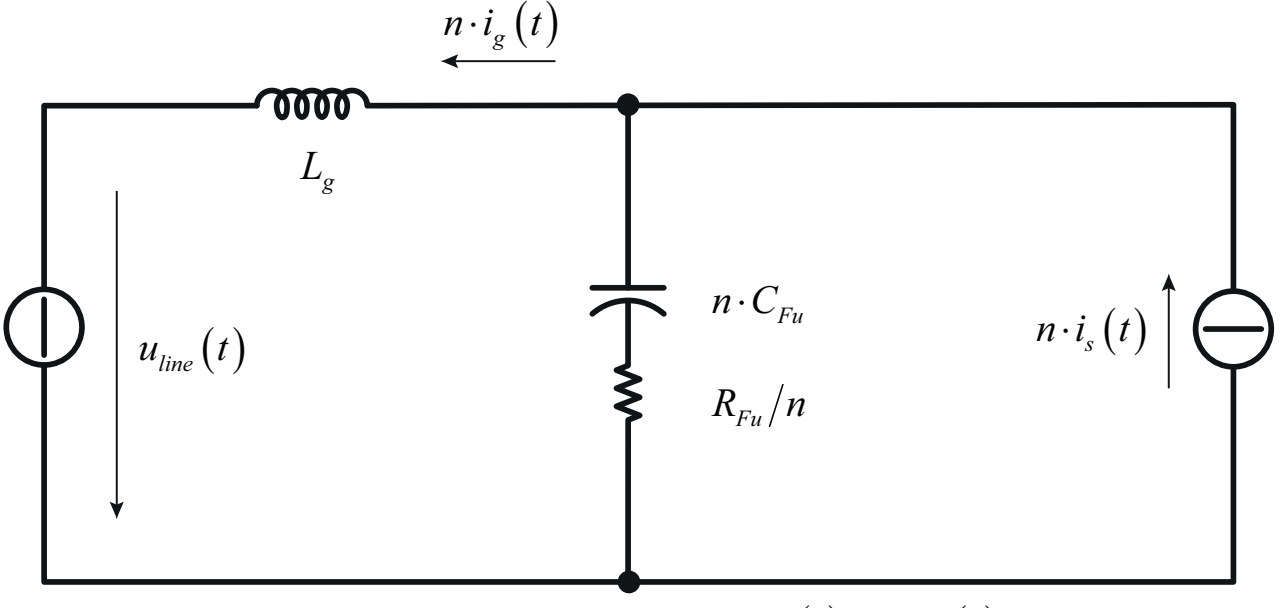

$$L_g = L_{line}(i_{cc}) + L_\sigma^{tr}(u_{cc}, P_n)$$

$$i_{line}(t) = n \cdot i_g(t)$$